Python for computational Problem Solving · Unit_04 · UNIT 4: Testing ,Debugging and Object Oriented Programming

DRAFT — UE26CS151A · Unit_04 · UNIT 4: Testing ,Debugging and Object Oriented Programming · Pytest, doctest, pdb

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Pytest, doctest, pdb

1 Pytest, doctest, pdb

Study focus: Pytest, doctest, pdb. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000014 · EV0000015

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

So first you create the file gedit file one.

Key details

- This particular statement will import the PDB.
- module in your program.
- That is PDB followed by dot followed by the function name that is set trace.
- If you give this statement, it will access the set trace method from the PDB module.
- In number, I am taking my number as 3, I am calling the function fact, I am going to pass the value as 3 because I want to calculate the factor of 3.
- and have to exit my program so control Q now I am going to give the command python followed by file name.py if you are using the second version give python followed by file name.py.
- If you are using the third version, you are supposed to give python 3 followed by file name.py.
- to print the value of a variable so first we need to create one program using g edit g edit followed by variable is my file name with the extension dot py.
- We know we are supposed to give the statement import followed by PDB.
- So next I'm use the variable A with the value 10.
- And how to trace we know we are supposed to give pdb dot set trace is a method that is available in the pdb.
- So this will trace each and every line what will happen in my program next I'm print my variable using the statement print and the variable is a and I will say my program control s control q is I will quit my program now I am going to execute we know how to execute the program python followed by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 05:00–09:59

variable dot p y so this will enter into my pdb debugger now if I want to print the value of the variable what is my variable have used in my program have used the variable a if I want to.

Key details

- No the value of that variable A I have to give the command P followed by variable name where the variable name in my program is A and you press enter you will get the value of that variable that is A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Pytest, doctest, pdb transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_04/AV_Summaries/001_Pytest_doctest_pdb__b3f1d7a13157fe7e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Pytest, doctest, pdb transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_04/AV_Summaries/001_Pytest_doctest_pdb__b3f1d7a13157fe7e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Pytest, doctest, pdb transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_04/AV_Summaries/001_Pytest_doctest_pdb__b3f1d7a13157fe7e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Pytest, doctest, pdb transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_04/AV_Summaries/001_Pytest_doctest_pdb__b3f1d7a13157fe7e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 10 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0021. Sidecar variants compare...
SRC00007	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0021. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0021. Sidecar variants compare...
SRC00009	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0021. Sidecar variants compare...

C Revision and retrieval practice